ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PRACTICE EXAM 4

MARKS: 80 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Paper 2 style (design-related, 80 marks). Section A: objective questions (20) · Section B: designing a robot headlight circuit and battery pack, prototyping on a breadboard, soldering and testing with a multimeter, and logic design for a simple robot alarm (60).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. Draw circuit diagrams neatly, using the correct symbols. A ruler and a non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The current-limiting resistor of an LED is connected …

A in series with the LED. B in parallel with the LED.
C across the battery only. D nowhere; it is not needed.

1.1.2 Four 1,5 V cells connected in series give …

A 1,5 V. B 3 V. C 6 V. D 4 V.

1.1.3 The E12 standard resistor value just above 140 Ω is …

A 140 Ω. B 150 Ω. C 145 Ω. D 135 Ω.

1.1.4 To measure current, a multimeter is connected …

A in parallel with the component. B in series with the component.
C directly across the battery. D to the earth wire only.

1.1.5 The solder used for electronics is mostly …

A tin (with lead or copper). B iron.
C aluminium. D zinc.

1.1.6 A dry (bad) solder joint looks …

A shiny and smooth. B perfectly cone-shaped.
C like a silver mirror. D dull and grainy.

1.1.7 The output of a 2-input OR gate is 0 when …

A any input is 1. B both inputs are 1.
C both inputs are 0. D the inputs are different.

1.1.8 The decimal digit 9 written in BCD is …

A 1001 B 1100 C 0110 D 1111

1.1.9 A pull-down resistor makes a logic input read … while its switch is open.

A HIGH B LOW C floating D 9 V

1.1.10 The long rows along the edges of a breadboard are used for …

A ICs only. B LEDs only.
C soldering. D the supply (power rails).

(10)

1.2 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.2.1–1.2.10).

	COLUMN A		**COLUMN B**
1.2.1	Protects a circuit from too much current	A	tilt switch
1.2.2	Standard range of resistor values	B	BCD
1.2.3	Opens and closes a circuit by hand	C	fuse
1.2.4	Group of cells connected together	D	breadboard
1.2.5	Poor solder joint that may not conduct	E	dry joint
1.2.6	Keeps a logic input LOW while its switch is open	F	battery
1.2.7	Measures voltage, current and resistance	G	pull-down resistor
1.2.8	Board for testing circuits without soldering	H	multimeter
1.2.9	Each decimal digit written as a 4-bit binary group	I	switch
1.2.10	Sensor that switches when a robot leans over	J	E12 series

(10)

TOTAL SECTION A: [20]

SECTION B

QUESTION 2: DESIGN A ROBOT HEADLIGHT CIRCUIT

2.1 A small robot needs two white LEDs as headlights. Each LED needs 3,2 V and 20 mA. The LEDs are connected in parallel, each with its own resistor, across a 6 V battery pack.

2.1.1 Calculate the value of each resistor and choose the nearest standard value above it. (3)

2.1.2 Calculate the total current drawn from the battery. (1)

2.1.3 Calculate the power dissipated in each resistor (use the calculated value). (2)

2.2 Draw a neat, labelled circuit diagram of the headlight circuit, with the battery pack, a fuse, an on/off switch and the two LED branches. (5)

2.3 The battery pack is made of four 1,5 V, 2 000 mAh AA cells in series. Calculate the voltage and capacity of the pack, and how long it can run the headlights. (4)

2.4 Explain why each LED should have its own resistor instead of sharing one resistor. (2)

2.5 Suggest a suitable fuse rating for this circuit and give a reason. (1)

2.6 Give TWO ways to identify the anode and the cathode of an LED. (2)

[20]

QUESTION 3: PROTOTYPING, SOLDERING AND TESTING

3.1 Describe how you would build the headlight circuit on a breadboard. Refer to the power rails and the rows. (4)

3.2 List, in order, FIVE steps to make a good solder joint on a circuit board. (5)

3.3 Describe what a good solder joint and a dry joint look like. (2)

3.4 Explain how you would use a multimeter to measure: (a) the battery voltage; (b) the current through one LED; (c) the value of a resistor. Give the meter setting and how it is connected in EACH case. (6)

3.5 One LED does not light up. Give THREE possible faults. (3)

[20]

QUESTION 4: LOGIC FOR A SIMPLE ROBOT ALARM

4.1 A robot's alarm (Y) must sound when its tilt switch T = 1, OR when the bumper switch B = 1 AND the alarm is armed (A = 1).

4.1.1 Write the Boolean expression for Y. (2)

4.1.2 Draw the truth table for Y. (4)

4.1.3 Draw the logic circuit for Y using AND and OR gates. (4)

4.2 The robot shows codes on a display. Convert: (a) the decimal number 7 to 4-bit binary; (b) the decimal number 39 to BCD. (4)

4.3 Give TWO reasons why digital systems use the binary number system. (2)

4.4 Draw the symbol and the truth table of a NOT gate. (2)

4.5 Explain why a pull-down resistor is used on a push-button input. (2)

[20]

TOTAL SECTION B: [60]

TOTAL: 80

END OF QUESTION PAPER